Fundamental and technical signals for A share selection

Investments 1005 | Group 9 | Semester 1, 2026

A cross-sectional stock-selection study using historical daily data from 2 January 2018 to 29 April 2026.

Research question and main finding

Can accounting and technical information improve net stock selection within the same opportunity set? Does processing the inputs or predictions add value beyond unprocessed scores and simple portfolios?

The main strategy uses Ridge forecasts and a Top-100 equal-weight portfolio. The 2021–2025 net return was 195.7%, with cumulative daily net excess +52.0 pp. In 2026 through April 29, Ridge returned 8.8% net, versus 8.2% for the matched benchmark. 2026 net daily excess was positive; the partial-year interval gives limited evidence about persistence.

The comparison retains all four methods, matched raw baselines and the same-cost equal-weight benchmark.

1. Adopted signals and investment rationale

Profitability, growth and financial-risk changes may predict relative returns or mean reversion. Price trends, volatility and activity may reflect information diffusion, reversal or liquidity. These are hypotheses, not guarantees.

The table lists the union of adopted inputs across annual fits. Each fold selects its inputs from mature historical labels; methods share one final opportunity set. Complete formulas are supplied in the data dictionary.

Input	Economic input / transformation
F1	Weighted ROE relative to its 60-day mean
F2	120-day volatility-adjusted change in four-report ROA trend
F3	120-day volatility-adjusted change in revenue-growth trend
F5	120-day volatility-adjusted change in debt-to-equity
T1	20-day momentum
T2	60-day momentum
T3	20-day return volatility
T5	20-day average turnover
F6	Signed book-to-market
F7	Signed trailing earnings yield
F8	Annualized ROA level
F9	Operating cash flow to revenue

2. Historical data and investable universe

The local historical snapshots cover 2,018 market sessions and 6,000 stock-ID slots. Daily prices, volume, turnover, ST status, listing age, suspensions, limit prices and accounting snapshots are aligned to a fixed security axis.

The main universe consists of Shanghai and Shenzhen A shares known on the signal date, excluding then-ST names, listings younger than six months and missing required inputs. All four selected methods, their unprocessed comparison and the benchmark use the final common opportunity set. Historical securities are not filtered by whether they survive today.

Raw snapshots contain 401,235 repeated report-slot observations with report or announcement dates later than the snapshot date. These are masked before feature construction. Date inconsistency does not prove every corresponding value leaked; original unrevised report vintages remain uncertified. Accounting composites require compatible report periods and valid denominators.

A forward close / previous-close chain reduces mechanical ex-right jumps and supplies consistent opening labels and daily valuation. It approximates company-action returns; ordinary cash-dividend arrival dates are not a separate account ledger. Later report revisions and vendor backfills remain data risks.

Before ranking, F1 values with absolute magnitude at most 1e-10 become zero. This fixed numerical correction removes floating residuals from economic ties; the formula, missing values, other features and larger values are preserved.

The model forecasts next-open to twenty-session opening returns. Each annual fit and factor selection uses only earlier labels whose exit has matured. The 2018-2020 model history is fitted, not an independent forecast evaluation. Models are compared using annual return consistency, same-cost excess, information ratio and drawdown.

3. Actual processing and model specifications

Recorded session counts: target horizon 20, rebalance cadence 20, training stride 5. The table shows input unions, processing and actual parameters. Annual selections are supplied separately.

Method	Frozen specification
Fundamentals	Input union F1, F2, F3, F6, F7, F8, F9; input EMA=0; score EMA=20; neutralization=industry + log market cap; fundamental blend weight=0%; construction=signed mean factor ranks; family weights=[1.0]; target=demeaned return; training years=expanding history.
Combined score	Input union F1, F2, F3, F5, T1, T2, T3, T5; input EMA=0; score EMA=40; neutralization=industry + log market cap; fundamental blend weight=0%; construction=signed mean factor ranks; family weights=[0.5, 0.5]; target=demeaned return; training years=expanding history.
Ridge	Input union F1, F2, F3, F5, T1, T2, T3, T5; input EMA=0; score EMA=20; neutralization=industry + log market cap; fundamental blend weight=0%; alpha=10000.0; target=demeaned return; training years=expanding history.
LightGBM	Input union F1, F2, F3, F5, T1, T2, T3, T5; input EMA=0; score EMA=40; neutralization=industry + log market cap; fundamental blend weight=0%; depth=3; leaves=7; min leaf=500; trees=56; learning rate=0.05; target=demeaned return; training years=expanding history.

EMA uses alpha=2/(span+1); zero bypasses it. First finite values initialize state; missing observations neither emit scores nor update state. Later fits do not replace past predictions.

Industry / log-market-cap residualization precedes score EMA and any fundamental rank blend. It does not guarantee style-neutral holdings, lower drawdown or improved returns.

4. Portfolio, costs and matched benchmark

The long-only Top 100 equal-weight targets refresh every 20 market sessions, continuously from the 2018-07-12 opening (offset 0; signal 2018-07-11). The calendar never restarts at year-end. Data, signals and valuation remain daily; blocked or unfunded orders retain daily retry rules.

Trades adjust portfolio differences. Suspensions and zero volume block trades; opening upper-limit buys and lower-limit sells are blocked. Unsold holdings remain, unsuccessful buys create no positions, and purchases respect T+1. Official quote-less suspensions carry their last mark until resumption or termination; the 60-day threshold is diagnostic, not a forced exit. A normal positive-volume day missing its close raises an error.

Equal weight in the identical historical complete-input A-share pool, with next-open executions every 20 trading session(s) on an offset-zero grid anchored at the original first trade, under identical trading restrictions and matched gross/net transaction costs.

Gross strategies compare with the gross benchmark and net strategies with the net benchmark. The benchmark shares the final historical eligible pool, execution filters and actual target-update cadence; changing the opportunity set requires recomputing the raw baseline on that same pool.

Research assumptions are 3 bps commission and 5 bps slippage per side. Sell stamp duty is 10 bps before 28 August 2023 and 5 bps thereafter. Minimum commission, round lots and small transfer fees are ignored. Daily bars cannot establish auction queue priority or precise market impact.

Unresolved formal delistings produce 26 main-run portfolio write-off events across 24 securities (52 gross/net ledger entries). Zero recovery is a conservative assumption, not proof of zero eventual recovery. Codes, dates and notional are recorded.

5. Corporate actions and illiquid claims

Specified share exchanges use final published ratios in the sourced event ledger. When old shares become a conversion entitlement, the account holds a nontradable receivable at the old holding's last market value. It enters net asset value but cannot finance purchases. At release, raw old shares convert at the final ratio into the successor security's current adjusted units.

A fixed interim mark is an accounting assumption. It avoids pretending shares trade before listing but suppresses unobserved claim-price variation and can concentrate a valuation change on release. It is not evidence that fair value remained constant or that the claim was risk free.

All holdings satisfying a sourced cash-option event are assumed to exercise on its final declaration date. This uniform rule is not an individual-account exercise log. Cash claims keep the old value until payment-day close, then take the cash-option amount; only the next market open can invest it. Administrative entitlement conversion is not charged simulated trading commission or stamp duty.

For AVIC, Yulong and Deppon, notices do not establish a precise investor-account credit date. The simulation uses the end of the fifth market session after the relevant completion announcement and permits investment next open. This settlement lag is an explicit assumption. Jingwei uses announced S+1 day-end credit, again investable next open.

corporate_actions.csv records event dates, final ratios, cash prices, official sources and assumptions. corporate_action_ledger.csv separately records recognition, payment and release for each method and cost book. Cash available for orders excludes pending claims even when net asset value includes them.

6. Four complete-history return curves

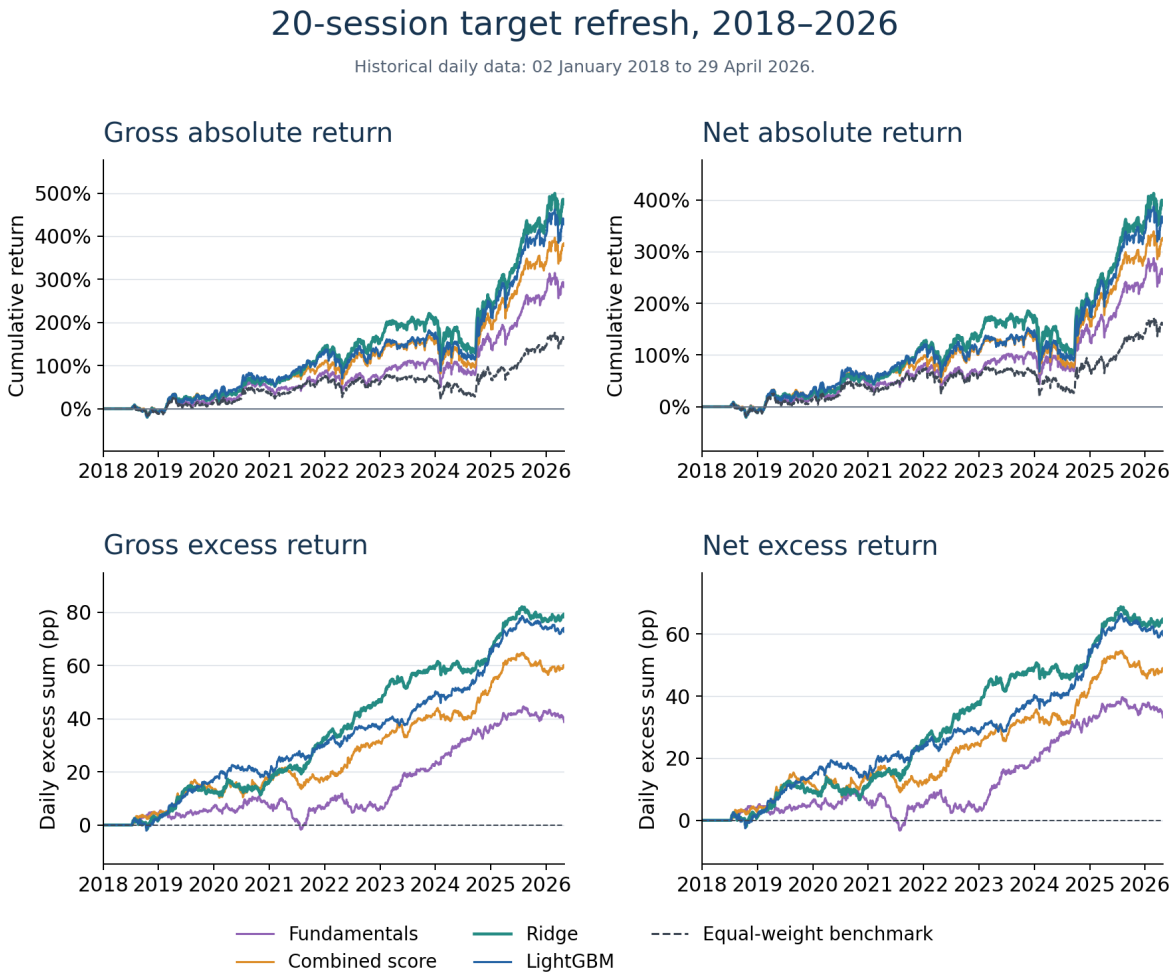


Figure 1. Historical backtest from 02 January 2018 to 29 April 2026, with 20-session target refresh and daily valuation.

Absolute return compounds daily returns: product of (1 + r) minus one. Excess return sums daily strategy minus same-cost benchmark returns, in percentage points; it is not the gap between compounded wealth curves.

7. 2021–2025 results

Excess sums same-cost daily differences in pp. Drawdown rebases initial wealth to one within the interval. Daily turnover is one-way traded weight.

Gross performance

Method	Return	Excess pp	Max DD	Daily turnover
Fundamentals	152.1%	+35.9	-30.3%	1.8%
Combined score	165.9%	+40.7	-32.5%	2.8%
Ridge	226.0%	+60.8	-32.8%	3.4%
LightGBM	204.8%	+55.8	-33.6%	3.2%
Equal-weight benchmark	78.3%	+0.0	-32.3%	0.3%

Net performance

Method	Return	Excess pp	Max DD	Daily turnover
Fundamentals	139.9%	+31.9	-30.3%	1.8%
Combined score	145.5%	+33.6	-32.6%	2.8%
Ridge	195.7%	+52.0	-33.0%	3.4%
LightGBM	178.4%	+47.7	-33.7%	3.2%
Equal-weight benchmark	76.6%	+0.0	-32.4%	0.3%

The 2021–2025 net return was 195.7%, with cumulative daily net excess +52.0 pp.

8. 2026 results to 29 April

Excess sums same-cost daily differences in pp. Drawdown rebases initial wealth to one within the interval. Daily turnover is one-way traded weight.

Gross performance

Method	Return	Excess pp	Max DD	Daily turnover
Fundamentals	3.8%	-4.0	-14.2%	1.9%
Combined score	10.2%	+1.7	-12.2%	2.7%
Ridge	9.4%	+1.1	-12.1%	3.5%
LightGBM	6.1%	-1.9	-13.8%	3.1%
Equal-weight benchmark	8.2%	+0.0	-13.3%	0.3%

Net performance

Method	Return	Excess pp	Max DD	Daily turnover
Fundamentals	3.5%	-4.3	-14.3%	1.9%
Combined score	9.7%	+1.3	-12.3%	2.7%
Ridge	8.8%	+0.6	-12.2%	3.4%
LightGBM	5.6%	-2.4	-13.9%	3.1%
Equal-weight benchmark	8.2%	+0.0	-13.3%	0.3%

In 2026 through April 29, Ridge returned 8.8% net, versus 8.2% for the matched benchmark. 2026 cumulative daily net excess was +0.6 pp, with maximum drawdown -12.2%.

2026 net daily excess was positive; the partial-year interval gives limited evidence about persistence.

9. Risk measures and yearly consistency

Net method / period	Ann. vol.	Zero-rf Sharpe	Info. ratio
2021–2025 / Fundamentals	24.6%	0.86	1.20
2021–2025 / Combined score	24.1%	0.89	1.20
2021–2025 / Ridge	23.9%	1.06	1.87
2021–2025 / LightGBM	25.3%	0.97	1.77
2021–2025 / Equal-weight benchmark	23.5%	0.62	n/a
2026 / Fundamentals	25.5%	0.58	-2.66
2026 / Combined score	22.2%	1.50	0.71
2026 / Ridge	23.4%	1.31	0.34
2026 / LightGBM	24.3%	0.86	-1.48
2026 / Equal-weight benchmark	23.5%	1.23	n/a

Sharpe assumes zero risk-free return. Information ratio annualizes daily excess mean / standard deviation with 252 sessions. The short 2026 record cannot establish persistence; 2026* is partial.

Year	Fund.	Combined	Ridge	LightGBM	Benchmark
2021	24.6%	24.7%	45.7%	38.2%	26.3%
2022	-10.3%	2.6%	2.4%	-5.8%	-9.5%
2023	25.3%	18.4%	20.1%	21.0%	7.5%
2024	18.2%	11.5%	7.3%	18.4%	3.8%
2025	45.0%	45.4%	53.7%	49.4%	38.4%
2026*	3.5%	9.7%	8.8%	5.6%	8.2%

10. Raw baseline and adopted designs

Raw and adopted portfolios share the final opportunity set and training calendar. The main curves show the adopted group; these tables retain its unprocessed counterpart.

2021–2025 / net performance

Method	Raw return	Adopted return	Raw excess pp	Adopted excess pp
Fundamentals	101.9%	139.9%	+16.1	+31.9
Combined score	139.9%	145.5%	+28.6	+33.6
Ridge	128.6%	195.7%	+23.2	+52.0
LightGBM	104.0%	178.4%	+15.6	+47.7

2026 to 29 April / net performance

Method	Raw return	Adopted return	Raw excess pp	Adopted excess pp
Fundamentals	13.3%	3.5%	+4.9	-4.3
Combined score	4.6%	9.7%	-3.6	+1.3
Ridge	2.9%	8.8%	-5.2	+0.6
LightGBM	8.4%	5.6%	+0.3	-2.4

Both groups use the same target-refresh cadence and opportunity set. Complete gross/net comparisons are supplied in experiment_metrics.csv.

Cadence comparison, same Ridge model: daily net returns are 253.1% in 2021–2025 and 6.0% in 2026; 20-session returns are 195.7% and 8.8%. The respective 2026 daily net-excess sums are -1.49 and +0.56 pp, each against its matching-cadence benchmark.

11. Interpretation and limits

Model risk. Repeated configuration comparisons can overfit a finite set of market regimes. Historical performance and neighboring parameter results do not establish a persistent return premium.

Data and implementation. Report revisions, label availability and complete-input filtering can bias samples. Daily limits, suspensions, conservative write-offs and approximate dividend accounting limit execution realism. Conversion claims use interim marks and three cash events use assumed settlement lags.

Style and model risk. The Top-N account can carry size, sector, volatility and market-beta exposures. A matched equal-weight benchmark controls the opportunity set but not all styles. Model relations can reverse during liquidity stress, speculative rallies, policy shifts or restatements.

Auxiliary reference	2026 price return	Ridge beta	Ann. intercept pp
CSI 500 price index	11.7%	0.82	-2.2

The auxiliary calculation is a zero-rf daily regression on the CSI 500 price index, not formal CAPM or a total-return benchmark. The annual intercept is 252 times the daily intercept; the common eligible-universe benchmark remains primary.

2026 net daily excess was positive; the partial-year interval gives limited evidence about persistence.

References and reproducibility

Bodie, Z., Kane, A., and Marcus, A. Investments, 13th edition. McGraw Hill. Portfolio risk and performance-evaluation framework.

Scikit-learn. Ridge documentation; time-series validation documentation. LightGBM. Parameters Tuning.

Ministry of Finance of the People's Republic of China. 2008 sell-side securities stamp-duty rule; Announcement No. 39 of 2023, effective 28 August. Commission and slippage are research assumptions.

The single supplementary ZIP contains prepared arrays, fitted models, source code, frozen research specifications, experiment metrics, event sources, daily returns, a data dictionary and running instructions. It supports model retraining and portfolio reconstruction from the prepared arrays. Rebuilding features from original financial statements requires access to the documented local raw database.

FourCharts_DailyValues.csv reproduces every chart value; Report_Metrics.csv reproduces the main stage and yearly statistics. research_selection.json identifies adopted processing, parameters, training cutoffs and source experiments. experiment_metrics.csv retains the raw baseline and actual attempted configurations. Corporate-action source URLs and settlement assumptions are recorded separately.